GG–G–91b
MAY 23, 1960
SUPERSEDING
Fed. Spec. GG–G–91a
July 14, 1953

FEDERAL SPECIFICATION

GAGE, TIRE PRESSURE, SELF CONTAINED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers direct-reading tire pressure gages, calibrated in pounds per square inch.

1.2 Classification.

1.2.1 *Types, classes, and styles.*—Tire pressure gages covered by this specification shall be of the following types, classes, and styles, as specified (see 6.2):

Type I.—Precision gage.
Type II.—General service gages.
  Class A.—Heavy Duty.
    Style 1.—Short extension, single-chuck foot.
    Style 2.—Long extension, dual-chuck foot.
  Class B.—Light Duty.
    Style 1.—Low pressure, single-chuck foot.
    Style 2.—High pressure, single-chuck foot.
    Style 3.—Air-Liquid, single-chuck foot.
Type III.—In-line chuck gages.
  Class A.—Replaceable cartridges.
    Style 1.—Single-chuck foot.
    Style 2.—Dual-chuck foot.
    Style 3.—Dual-chuck foot, standard and large bore.
  Class B.—Noncartridge.
    Style 1.—Single-chuck foot.
    Sytle 2.—Dual-chuck foot.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND OTHER PUBLICATIONS

2.1 Specifications, standards, and drawings.—The following specifications, standards, and drawings of the issues in effect on date of invitation for bids form a part of this specification:

Federal Specifications:
QQ–P–416—Plating, Cadmium (Electrodeposited).
ZZ–H–461—Hose; Gas (Acetylene-Hydrogen, Air and Oxygen).
PPP–B–585—Boxes, Wood, Wirebound.
PPP–B–591—Boxes, Fiberboard, Wood-Cleated.
PPP–B–601—Boxes, Wood, Cleated-plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Box, Fiberboard.
Federal Standards:
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking For Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of documents, U.S. Government Printing Office, Washington 25, D.C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:
MIL–P–116—Preservation, Methods of.
MIL–R–3065—Rubber, Fabricated Parts.
MIL–B–10377—Boxes, Wood, Cleated, Veneer, Paper Overlaid.

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Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection By Attributes.

MIL–STD–129—Marking for Shipment and Storage.

MIL–STD–130—Identification Marking of U.S. Military Property.

MS–33586—Metals, Definition of dissimilar.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Ordnance Corps Drawings:

8735758—Cartridge, Gage.

8735759—Valve, Cartridge.

8741502—Washer.

8741552—Washer, Fiber.

8741553—Washer.

(Copies of original size drawings will be furnished manufacturers upon request, by the procuring agency or as directed by that agency.)

2.2 Other publications.—The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

National Bureau of Standards Publication:

Handbook H28—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Preproduction sample.—When specified, the contractor shall furnish a complete gage sample for preproduction inspection in accordance with 4.2 (see 6.2). The preproduction sample shall be identical with the proposed production items. Unless otherwise specified, one sample shall be furnished (see 6.2). The approved sample and the preproduction report shall be used as the production standard.

3.2 Materials.—Materials shall be as specified herein. Materials not specifically designated shall be of good commercial quality. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Military Standard MS 33586.

3.2.1 *Castings.*—All castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, and other defects that may affect the structural strength or the designed serviceability of the casting involved. Strength and other essential physical and chemical properties of the castings shall be adequate to meet the performance requirements specified herein.

3.3 Construction.—The overall construction of the gages shall be free from any characteristics or defects that might render the gages unsuitable or inefficient for the intended purpose. All parts subject to maintenance disassembly shall be accessible for adjustment, repair, or cleaning without the necessity of special tools.

3.3.1 *Endurance.*—The gages shall be so constructed as to withstand not less than 2,000 actual gaging operations at their maximum operating pressure range, without the necessity of recalibration, adjustment, or repair. Gages shall be capable of withstanding a 25 percent ±5 percent overload, without damage, for a minimum of 10 minutes.

3.3.2 *Safety devices.*—Each tire gage shall be equipped with safety devices where practical. Parts which are hazardous to the operator shall be guarded.

3.3.3 *Shock.*—The tire gages covered by this specification shall be capable of withstanding, without damage, being dropped free fall from a minimum height of 3 feet to a concrete surface five times.

3.3.4 *Welding, brazing, and soldering.*—Welding, brazing, and soldering shall be performed in accordance with good commercial practice. In no event shall such processes, including peening or plugging, be used on castings or forgings for reclaiming any parts of the gages without authorization of the procuring activity.

3.3.5 *Fastening devices.*—All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts when subject to removal or adjusment shall not be swaged, peened, staked, or otherwise permanently deformed.

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3.3.6 *Threads.*—All pipe threads shall conform to the applicable requirements of the National Bureau of Standards Handbook H28. All other threads shall be in accordance with the tire gage manufacturer's commercial practice.

3.3.7 *Interchangeability.*—All replaceable parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model can be replaced or adjusted without requiring modification. Where practicable, all such parts shall be permanently and legibly marked with the manufacturer's part number in accordance with Military Standard MIL–STD–130.

3.3.8 *Metal finish.*—All functioning surfaces of the gages shall be smooth and free from tool marks. Fins, sprues, and other extra metal on castings and forgings shall be removed. All exposed sharp edges shall be broken with a radius or chamfer as applicable.

3.3.9 *Protective finish.*—All metal parts shall be of corrosion-resistant material, plated or treated to withstand the corrosive action of salt, fog, and high relative humidity as encountered in general service. Recalibration devices shall be of rust-resisting materials but not necessarily plated. Steel friction springs need not be plated. Any cadmium plating shall be accomplished in accordance with the requirements of Federal Specification QQ–P–416. Any special plating or treatment shall be as specified by the procuring activity (see 6.2). When specified (see 6.2), the finish shall be dull, dark in color, and nonreflective.

3.4 Design.—Tire pressure gages furnished under this specification shall be convenient to use, and entirely adequate in every respect for the intended purpose. Specific design characteristics shall be as specified herein for each type, class, and style.

3.4.1 *Illustrations.*

3.4.1.1 The illustrations shown in figures 1 through 7 herein are for the convenience of identification and are not intended to preclude the purchase of gages which are otherwise in accordance with the requirements of this specification.

3.4.1.2 Unless otherwise specified (see 6.2), the valve assembly, gaging element cartridge, and sealing washers for the type III, class A gages shall be in accordance with Ordnance Drawings 8735758, 8735759, 8741502, 8741552, and 8741553.

3.4.2 *Chuck foot.*—The chuck foot of all gages shall be offset and at an angle of 25 degrees to 35 degrees to the centerline of the gage barrel, except the lower chuck foot of type II, class A, style 2 gage which shall be offset and at an angle of 25 degrees to 35 degrees to the centerline of the gage barrel or a straight foot in line with the centerline of the gage barrel. The chucks shall be so constructed as to fit standard (0.302–32) threads used on tire valves. When specified (see 6.2), the chuck foot shall fit valves (0.482–26) threads used on off-road tires. When applied to the valve stem, either chuck shall depress the valve core plunger permitting the free passage of air from the tire into the gage pressure chamber, without damage to either the stem or core members. The chuck foot shall be made integral with the extension or connecting member, or rigidly attached thereto in such a manner as not to become loose or detached under normal usage. The types of chuck foot shall be as hereinafter defined for each type, class, and style of gage.

3.4.2.1 *Single-chuck foot.*—This type foot shall be provided with a single chuck which shall be so constructed as to operate the gage when depressed on the valve stem.

3.4.2.2 *Dual-chuck foot.*—This type of foot shall be provided with a dual opposed deflator pin. The dual-chuck foot assembly shall be so constructed as to operate the gage from one chuck and render the other chuck deflator pin inoperative. The deflator pins of type I and type II, class A, style 2 gages shall be hardened steel and shall be constructed so that distortion of the rubber sealing washer after long use cannot partially or completely block the flow of air through the gage.

3.4.3 *Types I and II gage.*—Gages of types I and II shall be composed primarily of a barrel with indicator bar and chuck foot with extension, except that type II, class B shall have no extension.

3.4.3.1 *Barrel.*—The barrel shall be so constructed as to fully enclose and shield the internal working mechanism. The inside of the

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barrel shall be treated to prevent corrosion, along its entire length.

3.4.3.2 *Indicator bar.*—The connection between the indicator bar and the internal actuating mechanism shall be such as to permit the indicator bar to remain at the indicated pressure after the source of pressure has been removed. The graduation marks shall be from 0.010 to 0.015 inch in width, but uniform on any one instrument, and from 0.004 to 0.008 inch in depth. The accuracy in graduation spacings shall be such as to meet with the requirements of each particular type and class. The numerals indicating the applicable graduation increment shall be a minimum 1/16 inch in height. Both graduations and numerals shall be clear cut and well defined to facilitate reading of the gage under normal service conditions.

3.4.3.3 *Type I, precision gage.*—Type I precision gages shall have a minimum pressure range of 10 p.s.i.g. to 160 p.s.i.g. and shall be 12 inches ±2 inches in overall length, and shall be provided with an extension 3½ to 4½ inches in length having a dual-chuck foot. The indicating bar shall be square in cross section with the graduations arranged progressively around all four sides. Gages of this type shall be similar in design to figure 1. Each type I precision gage shall be provided with an identifying sleeve covering the barrel of the gage so that the gage may be readily identified. The identifying sleeve shall be of polyethylene or other suitable plastic material, and shall be 6 inches ±½ inch in length. If plastic sleeves are utilized they shall not burn at a rate exceeding 2 inches per minute following ignition by a Bunsen burner. Plastic sleeves having a cellulose nitrate base are not acceptable. The ends of the sleeve shall be beveled for a distance of 1/8 to 3/16 inch with a 30-degree bevel to properly receive a beveled clamping ferrule. The sleeve shall be fluted or corrugated for handgrip. The beveled clamping ferrule shall be actuated by a knurled locking nut threaded on the shell or barrel of the gage so as to hold the identifying sleeve firmly in position on the shell, or barrel of the gage.

3.4.3.3.1 *Calibration.*—Precision gages shall be graduated in 1 p.s.i. increments from 10 p.s.i. up to 60 p.s.i. and in 5 p.s.i. increments from 60 p.s.i. to 160 p.s.i. with each individual increment being marked by a line and with each 10 p.s.i. increment marked with the appropriate numerical value. The graduated length of the scale shall be not less than 2½ nor more than 3½ inches.

3.4.3.3.2 *Accuracy.*—Precision gages, at any temperature from plus 70° F. down to minus 20° F. to minus 25° F., and within 5 seconds after pressure has been applied, shall indicate the true pressure within the following limits:

Accuracy at plus 70° F. ±5° F. (room temperature) to be within plus or minus
- 1 p.s.i. at 30 p.s.i.g.
- 1 p.s.i. at 50 p.s.i.g.
- 1.5 p.s.i. at 75 p.s.i.g.
- 2 p.s.i. at 120 p.s.i.g.

Accuracy at minus 20° F. to minus 25° F. to be within plus or minus
- 1 p.s.i. at 30 p.s.i.g.
- 1.5 p.s.i. at 50 p.s.i.g.
- 2 p.s.i. at 75 p.s.i.g.
- 3 p.s.i. at 120 p.s.i.g.

3.4.3.4 *Type II, general service gage.*—Type II general service gages shall be of two classes for general service usage and shall be furnished with flat, triangular, or square indicating bars, as specified herein.

3.4.3.4.1 *Type II, class A, heavy-duty general service gage.*—Gages of this type and class shall be of two styles and shall be of heavy construction throughout so as to withstand rough usage as encountered in service stations and tire shops. The type II, class A gages shall have a minimum graduated pressure scale range of 20 p.s.i.g. to 130 p.s.i.g. The gages of this type and class shall be graduated in not more than 2 p.s.i.g. increments up to at least 60 p.s.i.g. on the scale and in not more than 5 p.s.i.g. increments for the remainder of the scale. The scale shall have numerical markings at least every 10 p.s.i.g. intervals. The length of the graduated scale shall be not less than 3¼ inches (see figs. 2 and 3).

3.4.3.4.1.1 *Accuracy.*—Gages of this type shall indicate the true pressure, at any temperature from plus 70° F., down to minus 20° F. to minus 25° F., within 5 seconds after pressure has been applied, within the following limits:

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Accuracy at plus 70° F. ±5° F. (room temperature) to be within plus or minus
1.5 p.s.i. at 30 p.s.i.g.
1.5 p.s.i. at 50 p.s.i.g.
2 p.s.i. at 75 p.s.i.g.
3 p.s.i. at 120 p.s.i.g.
Accuracy at minus 20° F. to minus 25° F. to be within plus or minus
1.5 p.s.i. at 30 p.s.i.g.
2 p.s.i. at 50 p.s.i.g.
3 p.s.i. at 75 p.s.i.g.
3 p.s.i. at 120 p.s.i.g.

3.4.3.4.1.2 *Type II, class A, style 1, short-extension single-chuck foot gage.*—Gages of this type, class, and style shall be 8½ to 9½ inches in overall length, and shall be provided with a single-chuck foot with an extension 1½ to 2½ inches long (see figure 2).

3.4.3.4.1.3 *Type II, class A, style 2, long-extension dual-chuck foot gage.*—The gages of this type, class, and style shall be provided with a dual-chuck foot, an extension of not less than 5½ inches, and provisions to hang up the gages (see figure 3). Unless otherwise specified, the overall length of the gage (end of the gage barrel to the end of the chuck foot) shall be not less than 11 inches (see 6.2).

3.4.3.4.2 *Type II, class B, light-duty gage.*—Gages of this type and class shall be for light-duty, individual usage, and shall be provided with a pocket clip. The overall length of the gage shall be 5 to 6 inches.

3.4.3.4.2.1 *Accuracy.*—Type II, class B, style 1 and style 3 gages shall indicate the true pressure within plus 0 p.s.i. or minus 2 p.s.i. at plus 70° F. ±5° F. temperature and 50 p.s.i.g. pressure. Type II, class B, style 2 gages shall indicate the true pressure at plus 70° F. ±5° F. temperature within plus 1 p.s.i. or minus 2 p.s.i. at 75 p.s.i.g. reading, and within plus 3 p.s.i. or minus 3 p.s.i. at 100 p.s.i.g. reading. All type II, class B gages shall indicate the true pressure at minus 20° F. to minus 25° F. temperature within plus or minus 3 p.s.i. at 50 p.s.i.g. pressure. Gages shall indicate the true pressure at either of the above temperatures within the aforementioned limits within 5 seconds after the pressure has been applied.

3.4.3.4.2.2 *Type II, class B, style 1, low-pressure gage.*—Gages of this type class, and style shall have a minumum-pressure range from 10 to 50 p.s.i., graduated in 1 p.s.i. increments with the appropriate numeral at each mark on the indicator bars (see figure 4A).

3.4.3.4.2.3 *Type II, class B, style 2, high-pressure gage.*—Gages of this type, class, and style shall have a minimum-pressure range from 20 to 100 p.s.i. graduated in 5 p.s.i. increments maximum, with numerals at each 10 p.s.i. marking (see figure 4B).

3.4.3.4.2.4 *Type II, class B, style 3, air and liquid gage.*—Gages of this type, class, and style shall have a minimum-pressure range from 10 to 50 p.s.i., graduated and marked in 1 p.s.i. increments, and shall be fully protected against dry weight material, brine and other antifreeze solutions (see figure 4C). The indicator bar shall be attached to the internal mechanism so as to facilitate washing and oiling of the internal mechanism. A hole shall be provided in the outer end of the barrel to permit injection and ejection of a liquid cleaner and light oil.

3.4.4 *Type III, in-line chuck gage.*—Type III in-line gages shall be of two classes for incorporation into the air line, permitting air discharge into the tire by means of an inflator trigger or lever mechanism, and shall indicate the air pressure upon release of the trigger or lever. The lever or trigger shall be so located as to facilitate the operation of the inflation attachment when the operator is wearing heavy gloves or mittens. The lever or trigger shall be hinged at or near one end and shall develop a leverage ratio of not less than 3 to 1 on the air-valve assembly. All type III gages shall be constructed in such a manner as to be capable of withstanding a pressure of 150 p.s.i.g. without leakage. Gages of this type shall have a minimum pressure range from 10 to 120 p.s.i., with maximum graduations of 2 p.s.i. increments up to 40 p.s.i., and maximum graduations of 5 p.s.i. increments above 40 p.s.i. Each 10 p.s.i. increment shall be clearly marked with the numerical value. When specified, there shall be furnished a ¼-inch N.P.T. threaded connection with a 30-degree, female tapered seat (see 6.2).

3.4.4.1 *Accuracy.*—Accuracy requirements for type III gages shall be as follows: Construction shall be such as to insure accuracy within the tolerance of plus or minus ½ of 1

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percent of the full-scale reading at the 30 p.s.i. calibration, and within the tolerance of plus or minus 2 percent of the full-scale reading at the 80 p.s.i. calibration when subject to temperatures of from plus 110° F. to plus 115° F. Unless otherwise specified (see 6.2), at temperatures of minus 20° F. to minus 25° F., the accuracy shall be within the tolerance of plus or minus 1 percent of the full-scale reading at the 30 p.s.i. calibration, and within the tolerance of plus or minus 3 percent of the full-scale reading at the 80 p.s.i. calibration.

3.4.4.2 *Discharge hose, extension, and foot.*—Gages of type III shall be provided with a ¼-inch I.D. discharge hose not less than 14 inches in length and in accordance with tire gage manufacturer's commercial practice, except as otherwise specified (see 6.2). The hose shall be fitted with a metal extension having a dual-chuck foot (except that class A, and class B, style 1, shall have a single chuck) offset at an angle of 25 degrees to 35 degrees and so constructed as to fit all standard tire valves. The extension and the dual-chuck foot shall be 5½ to 6½ inches in overall length and shall swivel a full 360 degrees.

3.4.4.3 *Body.*—Gages of type III shall be provided with an opening on the air intake end threaded to receive a ¼-inch N.P.T. male fitting. A wire-cloth air filter, 40-mesh (0.010 inch), shall be furnished for the inlet opening. The air filter shall be readily accessible for cleaning. The body of the gage shall be of aluminum alloy or similar metal. When specified (see 6.2), there shall be furnished an additional dual-chuck foot with extension to fit large bore valves. Both of the dual footings shall be removable and replaceable at the end of the discharge hose by means of a ball-type, slip-sleeve, quick-acting coupler. Each of the dual footings shall be attached by chains with a ring to prevent the loss of either chuck or extension while not in use. The ring shall be constructed so as to slide freely over the discharge hose and shall be of such diameter as not to pass over the quick-acting coupler.

3.4.4.4 *Intake leader hose.*—When specified (see 6.2), there shall be furnished an intake leader hose to connect with the intake opening of the gage. The hose shall be in accordance with grade A, type II of Federal Specification ZZ–H–461, and 3 feet in length unless otherwise specified (see 6.2). The hose shall be ⅜-inch I.D. and fitted at one end with a male ¼-inch N.P.T. fitting, and on the other with a ⅜-inch hose-end universal coupler (see fig. 7). When specified (see 6.2), there shall be furnished a ¼-inch male N.P.T. threaded connection with a 30-degree female tapered seat.

3.4.4.5 *Type III, class A, replaceable cartridge in-line chuck gage.*—Type III, class A gages shall be furnished with a removable and replaceable gaging element enclosed in a self-contained cartridge, and the valve mechanism shall be removable and replaceable as a unit. To facilitate repair in the field, both the gaging element and the valve unit shall be removable and replaceable without the use of special tools. The type III, class A gage valve assembly, gaging element, and sealing washers shall comply with paragraph 3.4.1.2.

3.4.4.5.1 *Type III, class A, style 1, in-line single-chuck foot gage.*—Gages of this type, class, and style shall be furnished with a single-chuck foot.

3.4.4.5.2 *Type III, class A, style 2, in-line dual-chuck foot gage.*—Gages of this type, class, and style shall be identical with type III, class A, style 1, except that they shall have a dual-chuck foot with extension 5½ to 6½ inches in length (see fig. 5).

3.4.4.5.3 *Type III, class A, style 3, in-line dual-chuck foot gage.*—Gages of this type, class, and style shall be furnished with an additional dual-chuck foot with extension to fit large-bore tire valves. Both of the dual footings shall be removable and replaceable at the end of the discharge hose by means of a ball-type slip-sleeve quick-acting coupler. Each of the dual footings shall be attached by a chain with rings to prevent the loss of either chuck while not in use. The ring shall be so constructed as to slide freely over the discharge hose and shall be of such diameter as not to pass over the quick-acting coupler (see figure 5A). An intake leader hose shall be furnished when specified (see 6.2).

3.4.4.6 *Type III, class B, noncartridge, in-line chuck-gage.*—Type III, class B gages shall be two styles and shall be similar to the type

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III, class A gage, except that they shall not be required to have removable valve and gaging cartridges.

3.4.4.6.1 *Type III, class B, style 1, in-line chuck gage.*—Gages of this type, class, and style shall have a single-chuck foot.

3.4.4.6.2 *Type III, class B, style 2, in-line chuck gage.*—Gages of this type, class, and style shall be identical with type III, class B, style 1 gage, except that they shall have a dual-chuck foot with an extension 5½ to 6½ inches in length (see figure 6).

3.5 Storage temperatures.—All gages and all components thereof shall be able to withstand without damage storage in a temperature of minus 65° F. for periods up to 72 hours, and a 24-hour cycle of temperature, starting at 10 hours at plus 90° F., followed by 5 hours of steady increase to plus 155° F., 4 hours at plus 155° F., and 5 hours of steady decrease to plus 90° F. Following these exposures, and after being stabilized at a temperature of plus 70° F. ±5° F., any gage of any type, class, or style, shall be capable of meeting the applicable accuracy and performance requirements set forth in this specification.

3.6 Identification of product.—Each tire gage, except as noted below, shall be permanently and legibly marked to give the following desired information: Manufacturer's name or trademark and U.S. Property Stock Number. In the case of type II, class B gages, there shall only appear the manufacturer's name or known trademark. In the case of type III gages, there shall appear a serial number, as well as the manufacturer's trademark and U.S. Property Stock Number when specified. On the box or container for each gage, there shall appear the following information:

Gage Tire Pressure_____
Type _____
Class _____
Style _____
Fed Spec No. GG–G–91b_____
Mfr. Model No._____
Order No_____
Stock No._____
(Manufacturer's name or trademark U.S. property)_____

3.6.1 *Manufacturer's model number.*—The manufacturer's model number, marked in the applicable space provided, shall be identical with the manufacturer's engineering production.

3.7 Instruction leaflet.—Each type II, class B, style 3, and type III, classes A and B tire gage shall be furnished with a leaflet covering care and operation. The leaflet shall be that normally furnished by the contractor in general commercial practice.

3.8 Workmanship.—Workmanship shall be first class to assure procurement of good quality equipment of a neat general appearance. Standards of workmanship shall be such that equipment specified herein shall comply with the requirements in section 3. All components, subassemblies, parts, and the final assembly shall comply with tests in section 4.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Preproduction inspection. — Unless otherwise specified, preproduction inspection shall consist of all the tests specified in 4.4 and examination in accordance with 4.3.2 except that sampling in accordance with Military Standard MIL–STD–105 shall not apply (see 6.2). Preproduction samples shall be as specified in 3.1. Unless otherwise specified, preproduction inspection shall be performed by the contractor (see 6.2). Failure of preproduction sample to pass preproduction inspection shall be cause for rejection.

4.3 Acceptance inspection.—Acceptance inspection shall consist of the examination specified in 4.3.2 and the tests specified in 4.3.3.

4.3.1 *Acceptance lot.*—Acceptance lot shall consist of all gages of one type, class, and style

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from one manufacturer submitted for acceptance.

4.3.2 *Acceptance examination.*—The gages shall be examined in accordance with Military Standard MIL–STD–105 using the following classification of defects and acceptable quality level (AQL):

Classification of defects.

Major Defects

(AQL shall be 2.5 percent defective)

1. Defective casting affecting operation (see 3.2.1).
2. Protective finish not as specified (see 3.3.9).
3. Chuck foot not as specified (see 3.4.2).
4. Dual-chuck foot not as specified (see 3.4.2.2).
5. Indicator bar not as specified (see 3.4.3.2, 3.4.3.3, 3.4.3.4, 3.4.3.4.2.2, and 3.4.-3.4.2.4).
6. Pressure range not as specified (see 3.4.3.3, 3.4.3.4.1, 3.4.3.4.2.2, 3.4.3.4.2.3, 3.4.3.4.2.4, and 3.4.4.).
7. Indicator bar markings not as specified (see 3.4.3.2, 3.4.3.3, 3.4.3.3.1, 3.4.3.4.1, 3.4.3.4.2.2, 3.4.3.4.2.3, 3.4.3.4.2.4, and 3.4.4).
8. Discharge hose assembly not as specified (see 3.4.4.2).
9. Air filter not as specified (see 3.4.4.3).
10. Intake leader hose assembly not as specified (see 3.4.4.4).

Minor Defects

(AQL shall be 4.0 percent defective)

51. Fastening devices not as specified (see 3.3.5).
52. Gages with tool marks or rough finish (see 3.3.8).
53. Identifying sleeve not as specified (see 3.4.3.3).
54. Dimensions not as specified (see 3.4.3.3, 3.4.3.4.1.2, 3.4.3.4.1.3, and 3.4.3.4.2).
55. Lever or trigger inoperative (see 3.4.4).
56. Identification marking illegible (see 3.6).
57. Instruction leaflet not as specified (see 3.7).

4.3.3 *Acceptance tests.*—From each lot which has passed the acceptance examination specified in 4.3.2, samples for acceptance tests shall be selected in accordance with inspection level L5 of Military Standard MIL–STD–105. The samples selected shall be subjected to the accuracy test (see 4.4.1). Acceptance or rejection shall be based on an AQL of 4.0 percent defective.

4.4 Tests.

4.4.1 *Accuracy test.*—Gages shall be tested at a temperature of plus 70° F. ± 5° F. and shall indicate the true pressure within the limits of accuracy required of the particular type, class, and style of gage being tested in relation to the temperature referenced above. A standard air pressure measuring device having an overall accuracy of one-fourth of 1 percent of full-scale value shall be used as a criterion (see 3.4.3.3.2, 3.4.3.4.1.1, 3.4.3.4.2.1, and 3.4.4.1).

4.4.2 *Overload test.*—Each type, class, and style of gage shall be subjected to an overload of 25 percent ±5 percent in excess of its rated capacity for a minimum period of 10 minutes. The gage shall then be subjected to the applicable operational test as outlined in 4.4.1 (see 3.3.1).

4.4.3 *High and low temperature operational test.*—Gages shall be tested at the temperatures and air pressures specified in the applicable requirement for accuracy and shall indicate the true pressure within the limits of accuracy required for the particular type, class, and style of gage being tested. A standard air pressure measuring device having an overall accuracy of one-fourth of 1 percent of full-scale value shall be used as a criterion. All type I and type II gages shall indicate the true pressure within the specified tolerances at the applicable temperatures and pressure within 5 seconds after pressure has been applied (see 3.4.3.3.2, 3.4.3.4.1.1, 3.4.3.4.2.1., and 3.4.4.1).

4.4.4 *Storage test.*

4.4.4.1 *Cold test.*—The gage shall be placed in an ambient air temperature of minus 60° F. to minus 65° F. for a period of 72 hours. At the end of this period the gage shall be allowed to stabilize at an air temperature of 70° F. ±5° F. The gage shall then be inspected for evidence of damage and tested in accordance with 4.4.1 (see 3.5).

4.4.4.2 *Heat test.*—The gage shall be subjected to a 24-hour cycle of temperature beginning with 10 hours at plus 90° F. ±5° F., fol-

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lowed by 5 hours of steady increase to plus 150° F. to 155° F., 4 hours at plus 150° F. to 155° F., and 5 hours of steady decrease to plus 70° F. ±5° F. The gage shall then be inspected for evidence of damage and tested in accordance with 4.4.1 (see 3.5).

4.4.5 *Endurance test.*—The types I and II gages shall be actuated by the application of air pressure and the indicator bar closed after each application of air, for not less than 2,000 gaging operations or cycles. The type III gage shall be actuated by the application of air pressure and the lever or trigger operated so as to return the gage to zero after each application of air, for not less than 2,000 gaging operations or cycles. The type I and type II, class A gages shall be operated from an air source of 150 to 160 p.s.i.g. The type II, class B, styles 1 and 3 gages shall be operated from an air source of 45 to 50 p.s.i.g. The type II, class B, style 2 gage shall be operated from an air source of 90 to 100 p.s.i.g. The type III gages shall be operated from an air source of 115 to 120 p.s.i.g. and the lever or trigger shall be actuated not less than 2,000 times to assimilate the required number of gaging operations. The chuck foot on the type III gages shall be applied to the air source at least 1,000 times on each side of the chuck foot for a total of 2,000 gaging operations or cycles. The gaging operations or cycles shall be at a rate of approximately 10 per minute for the types I, II, and III gages. The method of applying air to the gages, actuating the indicator bar on the types I and II gages, and actuating the lever or trigger on the type III gages is at the discretion of the testing facility. Failure of any gage to pass this test in conformance with 3.3.1 shall be cause for rejection.

4.4.6 *Shock test.*—The tire gage shall be dropped from a vertical distance of 3 feet onto a concrete surface five times. The gage shall be examined for damage and the accuracy shall be tested in accordance with 4.4.1 to determine compliance with 3.3.3. Failure of any gage to pass this test shall be cause for rejection.

4.4.7 *Flammability test.*—The sleeve for test shall be clamped in a support with the sleeve extended outward. Under the sleeve there shall be clamped a piece of 10-mesh Bunsen burner gauze about 5 inches square, in a horizontal position, ¼ inch below the bottom of the sleeve, and with about ½ inch of the sleeve extending beyond the edge of the gauze. A Bunsen burner with an efficient flame ½ to ¾ inch in height, shall be adjusted so that the flame tip will just contact the specimen when placed under the free end of the sleeve. At the end of 30 seconds the flame shall be removed and the sample allowed to burn. The rate of travel of the flame shall be determined from the time required for the flame to travel the length of the sleeve after the removal of the Bunsen burner. In case the sleeve does not continue to burn after the first ignition, the burner shall be replaced under the free end for a period of 30 seconds immediately following the extinction of the flame. If the sleeve does not continue to burn until the flame has reached the end of the sleeve after the second ignition, the sample shall be considered self-extinguishing. When tested as described, the plastic sleeves shall not burn at a rate exceeding 2 inches per minute (see 3.4.3.3). Failure of the gage to pass this test shall be cause for rejection.

4.5 Inspection of preparation for delivery.—The preservation, packaging, packing, and marking of the gages shall be examined and tested to determine conformance with the requirements of section 5 of this specification, and as required by specifications referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with Military Standard MIL–STD–105, with inspection level L–7, and an AQL of 4.0 percent defective.

4.6 Defective units.—Units found defective in the course of inspection shall be corrected to the satisfaction of the Government or shall be replaced with nondefective units.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Federal Standard No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.—Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

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5.1.1.1 *Cleaning and drying.*—The gages shall be cleaned in accordance with process C–1 and dried in accordance with procedure D–1 or D–4 of Military Specification MIL–P–116.

5.1.1.2 *Preservation.*—Each gage shall be individually preserved in accordance with method IA of Military Specification MIL–P–116.

5.1.1.3 *Intermediate packaging.*—Identical gages shall be intermediate packaged in fiber boxes conforming to class 2 of Federal Specification PPP–B–636. The boxes shall be closed in accordance with the appendix to the box specification. The gross weight of intermediate packages shall not exceed approximately 20 pounds.

5.1.2 *Level C.*—The gages shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and physical damage during shipment to first receiving activity.

5.2 Packing.—Packing shall be level A, B, C, or D, as specified (see 6.2).

5.2.1 *Level A.*—The preserved and packaged gages shall be packed in boxes conforming to Federal Specifications PPP–B–585 (class 3), PPP–B–591 (overseas type), PPP–B–601 (overseas type), PPP–B–621 (class 2), PPP–B–636 (class 2 or 3), or Military Specification MIL–B–10377 (overseas type). The boxes shall be closed and strapped in accordance with the applicable specification or the appendix thereto. Unless otherwise specified, gross weight for wood and wood-cleated boxes shall not exceed 200 pounds (see 6.2). Gross weight for boxes conforming to Federal Specification PPP–B–636 shall not exceed the weight limitations of the box specification.

5.2.2 *Level B.*—The preserved and packaged gages shall be packed in boxes conforming to Federal Specifications PPP–B–585 (class 1), PPP–B–591 (domestic type), PPP–B–601 (domestic type), PPP–B–621 (class 1), PPP–B–636 (class 1 or 2), or Military Specification MIL–B–10377 (domestic type). Boxes shall be closed and strapped in accordance with the applicable box specification or the appendix thereto.

5.2.3 *Level C.*—The gages shall be packed to insure carrier acceptance and to afford protection against damage during direct shipment from supply source to first receiving activity.

5.2.4 *Level D.*—For Civil agencies only, the gages as specified in 5.1.2 shall be packed according to manufacturer's commercial export practice to insure acceptance by carrier and safe delivery to destination at the lowest rate or rating. Containers shall comply with carrier rules and regulations applicable to the mode of transportation without assessment or penalty charges for improper packing.

5.3 Marking.

5.3.1 *Military agencies.*—In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

5.3.2 *Civil agencies.*—Interior packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

6. NOTES

6.1 Intended use.—The gages covered by by this specification shall be suitable in every respect for measuring or checking the pressure of air in tires used on motor vehicles in Government service.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title number, and date of this specification.
(*b*) Type, class, and style required (see 1.2).
(*c*) Preproduction inspection when required and sample quantity, if different (see 3.1).
(*d*) Dull, dark, and nonreflective finish, when required (see 3.3.9).
(*e*) Special plating or treatment when required (see 3.3.9).
(*f*) Replaceable gaging elements and washers if different (see 3.4.1.2).
(*g*) Different chuck foot size when required (see 3.4.2).
(*h*) Gage overall length if different (see 3.4.3.4.1.3).
(*i*) Threaded connection when required (see 3.4.4 and 3.4.4.4).
(*j*) Gage accuracy at low temperature if different (see 3.4.4.1).

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(*k*) Discharge hose length if different (see 3.4.4.2).
(*l*) Large bore dual-chuck foot with extension when required (see 3.4.4.3).
(*m*) Intake leader hose when required (see 3.4.4.4 and 3.4.4.5.3).
(*n*) Intake leader hose length if different, and threaded connections when required (see 3.4.4.4).
(*o*) Intake leader hose for type III, class A, style 3, when required (see 3.4.4.5.3).
(*p*) Preproduction inspection if different (see 4.2).
(*q*) Responsibility for preproduction inspection if different (see 4.2).
(*r*) Level of preservation and packaging, level of packing required (see 5.1 and 5.2).
(*s*) Gross weight if different (see 5.2.1).

6.3 Transportation description.—Transportation description applicable to these items is:

Gages, air, not otherwise indexed by name.
Carload weight 24,000 pounds subject to Rule 34
Uniform Freight Classification, Truckload weight 24,000 pounds.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

CUSTODIAN:
Army—Ordnance Corps
Navy—Bureau of Yards and Docks
Air Force—Topeka
Other Interest:
Navy—Sh

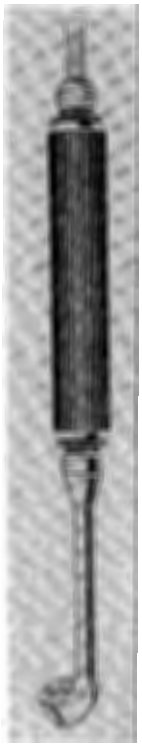


FIGURE 1.—Type I precision gage.

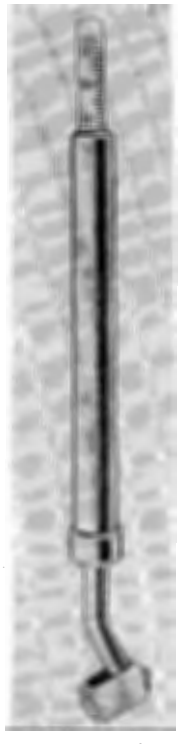


FIGURE 2.—Type II, class A, style 1 general service gage.

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FIGURE 3.—Type II, class A, style 2 general service gage.

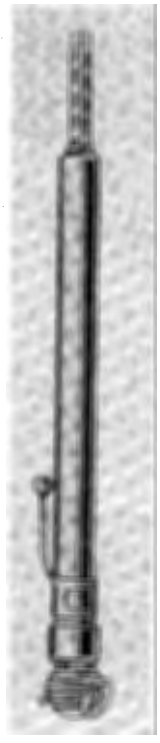


FIGURE 4A.—Type II, class B, style 1 general service gage.

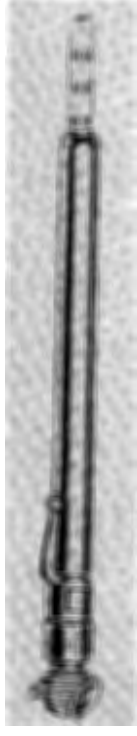


FIGURE 4B.—Type II, class B, style 2 general service gage.

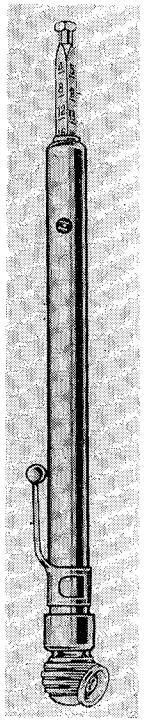


FIGURE 4C.—Type II, class B, style 3 general service gage.

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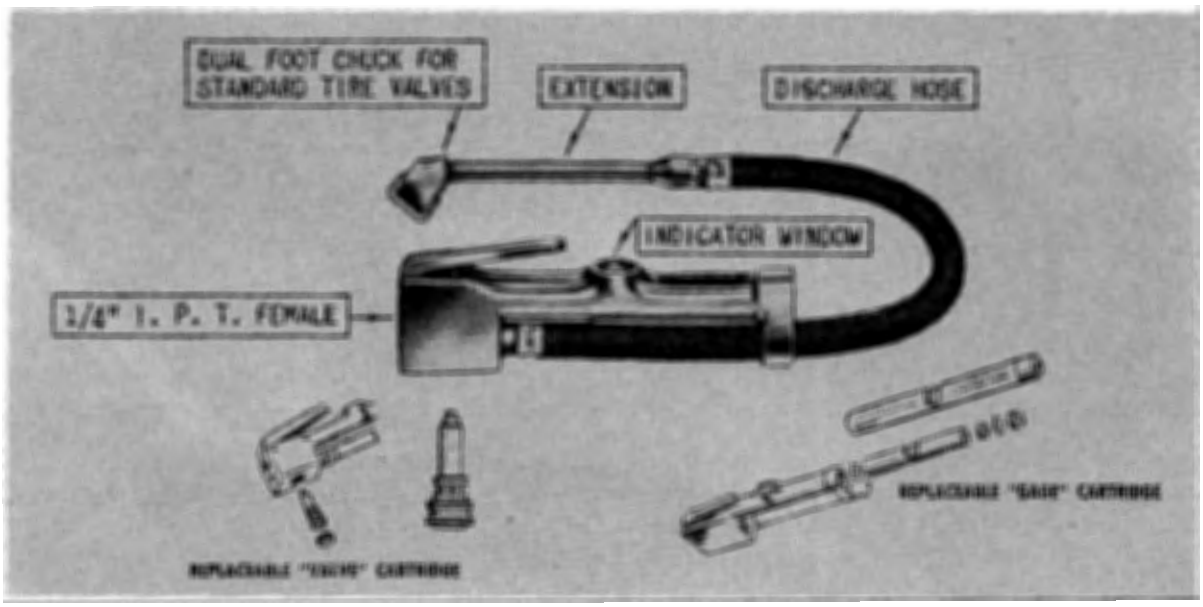


FIGURE 5.—Type III, class A, style 2 in-line chuck gage.

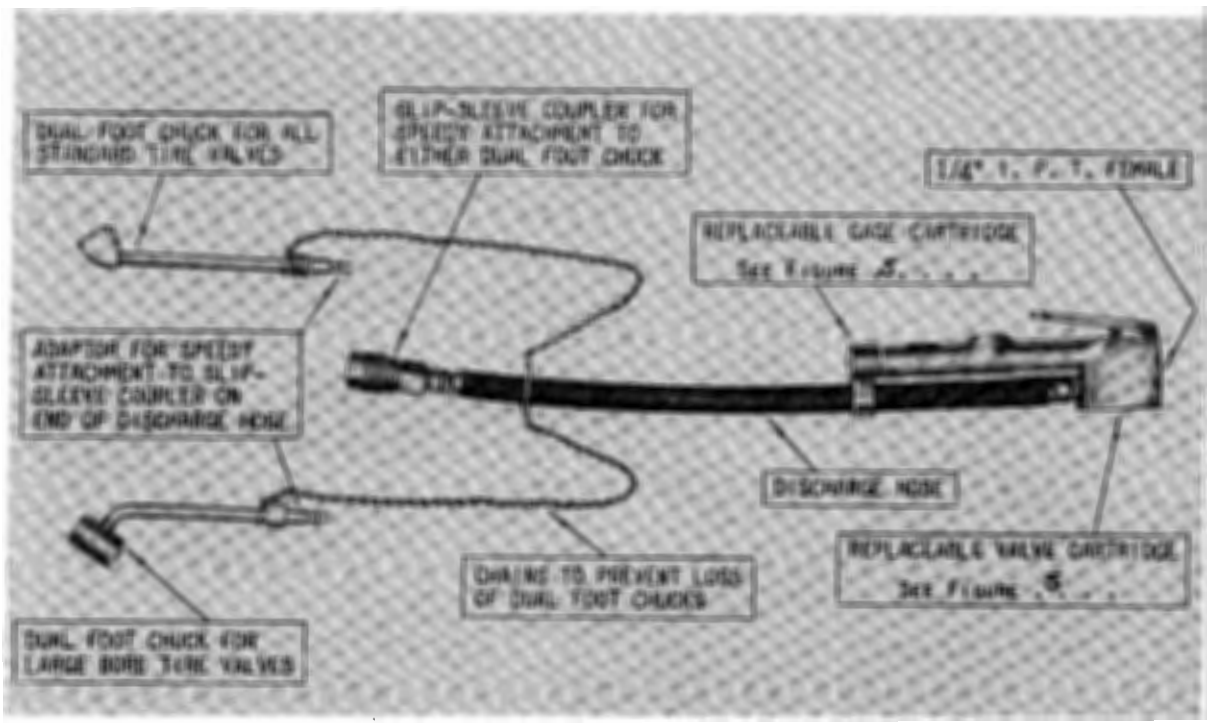


FIGURE 5A.—Type III, class A, style 3 in-line chuck gage.

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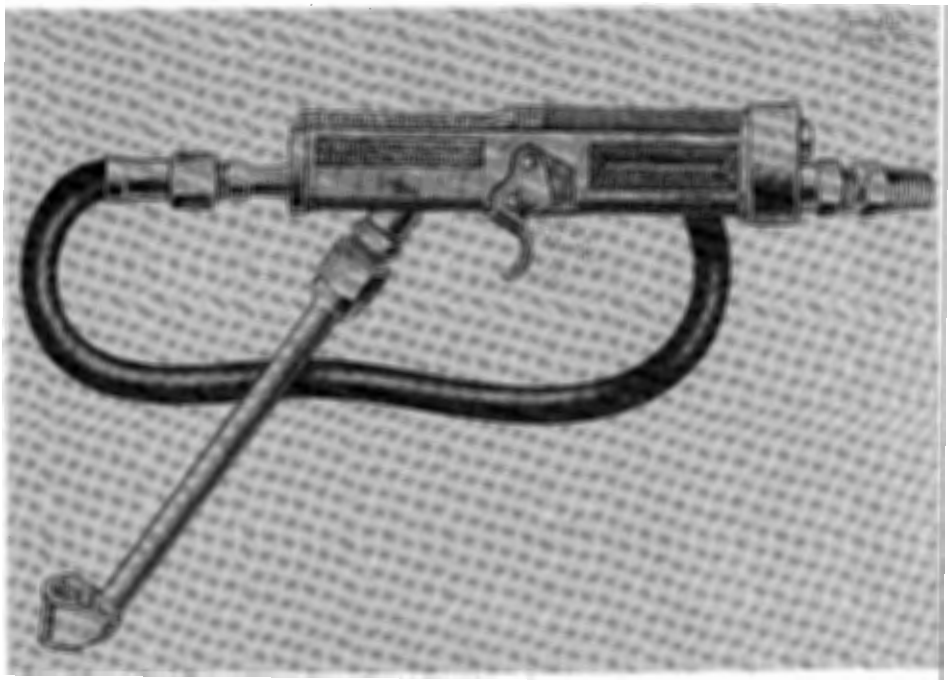


FIGURE 6.—Type III, class B, style 2 in-line chuck gage.

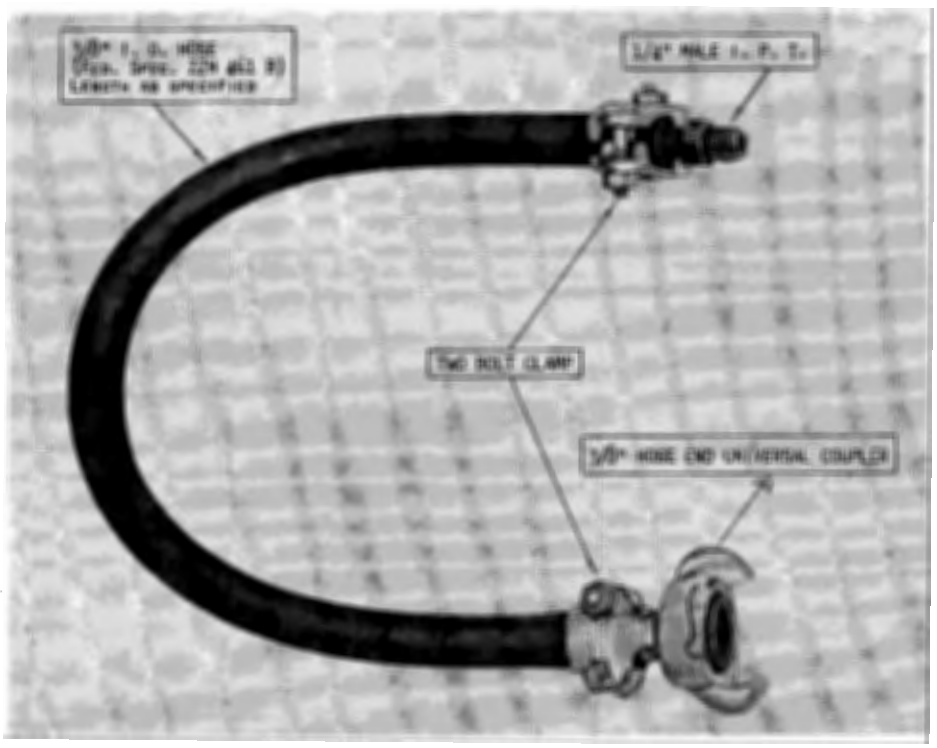


FIGURE 7.—Intake leader hose.

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